

The Decision I Need

One fork, the argument for it, and the counter-argument expected

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Update, 23 August. This document's argument stands, but its numbers are from an earlier stage; `06-status-report.pdf` carries the current ones. Since this was written: the population count went from 18 to 47; the direction rule passed a sealed test committed before the data existed (**9 of 10** never-measured states, best constant 6); the theory's own regime prediction was confirmed 27 of 27, nine pre-registered; and a second sealed campaign then showed the fixed 0.54 tipping point is population-dependent (located values 0.28–0.65), which narrowed the claim honestly rather than breaking it. The fork below is unchanged — if anything, the sealed pass strengthens the "stands on its own" branch. **Further update, 25 August.** Now 57 populations. The paper has since been through three adversarial review panels — twenty-four independent reviews, no rejects — and every cheap fix they converged on is applied: the deployable (deterministic) form of the mitigation preserves the predicted direction on 10 of 10 populations, the "lottery" finding is bounded to severe operating points, a 2022 anomaly was traced to the nominal income label sliding under inflation and resolved, and all eleven registered tests now have their registration and scoring commits pinned in the paper. The paper is 15 pages pending your page-limit answer, which gates the planned compression. The fork below still stands as written.

The question

A 2026 theory paper proves the effect can go either way, with no experiments in it. I have the experiments, a rule you can apply, and — since I last wrote this — a result that says the theory is right about one half of its claim and wrong about the other. Is that an empirical paper worth publishing, or does it read as a follow-up to someone else's result?

Why it is a genuine fork

If the empirical half stands on its own, the paper is close to ready. Their paper proves the possibility and contains zero experiments. Mine has 18 independent populations across five domains, two countries and two decades; a controlled design that separates the rate from the difficulty of the task; and a rule computable from a historical approval rate.

The strongest thing I have is a two-sided result about their theory. They split the world into two regimes — whether the protected attribute is available when the decision is made.

- Their **conditions**, the ordering relations their theorem is stated over, **cannot be evaluated on any of my 26 arms** — the quantity diverges on real data, so the two regions' ranges

always overlap. They are *sufficient, not necessary*, so this says their theorem is **silent** on real populations, not that it is wrong. A relaxed form of the same ordering tracks the direction on 24 of 26.

- Their **regime distinction** holds on **18 of 18**. In the attribute-aware regime they predict the direction is determined rather than variable, and it is: the advantaged group loses and the disadvantaged gains in every population, with no exceptions.

So I am not applying their theory and not simply refuting it. I am showing which half survives contact with data. That is what an empirical paper can settle and a population-level argument cannot.

If it now reads as **derivative**, that is a problem of framing rather than content, but it is still a problem. I reached this before knowing their paper existed — provable from dates — but a reader encountering both will see theirs first.

What I would argue

The "when" is the contribution, and it stands on its own:

1. **Theory says it *can* go either way. Only measurement says *when*.** That is the gap, and it is the gap a practitioner is standing in.
2. **Their conditions do not work on data.** Zero of 26. That is a finding only someone with the experiments could have produced, and it is a contribution against the theory rather than an application of it.
3. **My rule is computable and theirs is not.** Theirs needs the joint distribution of the model's scores and group membership. Mine needs last year's approval rate.
4. **The practical reading is sharper than I first thought, and less comfortable.** I originally claimed loans, hiring and admissions are nearly all stingy systems, so the harmful direction is the usual one. **I measured it, and that was wrong** — they span the whole range and sit on both sides of the switch. The real practical claim is better: two products in the *same* mortgage market, run through the *same* fairness constraint, move in opposite directions, and the fairness report says success in both cases.

The objection I expect

That this is now the empirical appendix to someone else's theory paper.

My answer: their conditions fail on every dataset I have, so this is not an application of their result — it is a demonstration that their result cannot be applied, plus a substitute that can. But I would rather be told now if that is too fine a distinction to carry a paper.

Other questions, if there is time

1. **Which venue, and when?** The deadline decides the page limit, which decides how much of **34 research documents** survives into the paper. My default is FAccT or AIES; there is no new algorithm here and the theory content is thin, which is a description of the work rather than a complaint about it.
2. ~~One missing comparison.~~ **Done, and it failed.** I ran the post-processing version across 17 populations: the rule vanishes, $r = -0.024$ against $+0.585$ for my own method. That is the regime boundary above rather than fragility — post-processing reads the protected attribute, which is the regime where the theory says nothing varies. But it means **every claim in the**

paper is now explicitly about attribute-blind in-processing, which is where almost every deployed system sits, because per-group thresholds are disparate treatment on their face. Is narrowing it that way a strength or a weakness in your reading?

3. **AI assistance.** I used an AI assistant substantially — for writing code, running analyses and drafting. What does the department expect me to disclose, and how?

What is not in question

The measurements. Every headline number is regenerated from stored results by an automated check that fails if a document and its data disagree — 28 such checks at the time of writing.

What I would want you to press me on

I withdrew two headline results this week. My strongest correlations, $+0.979$ and $+0.968$, came from arms where the model was worse than always predicting the majority label. I found it because a dataset where 89% of people pass forced the issue, and the same fix then rescued the mortgage result that had been failing. The method that caught it — pre-registering every prediction with the naive alternative it has to beat — is the part of this work I would most like judged, because it is also the part that keeps producing results I did not want.

Three of my own predictions failed and are written up as failures. Two of my own claims were withdrawn. I would rather be told the framing is wrong now than find out in review — which is exactly what happened with the remedy, and why I no longer claim it as new.